

Manual

Monitoring of Shop Floor Data BDE-MAB 8.2

Version 1.0.23570

Last change on: 08.10.2020

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1 Overview – Shop Floor Data / Order Data Monitoring

Overview

Purpose

The shop floor data / order data monitoring function provides numerous functions and overviews to monitor the orders / operations existing in the system. For this, the master data of the order / operation are accessed and compared with recorded data to provide an overview of the progress.

Implementation Notes

The shop floor data / order data monitoring function is used when

- You wish to obtain an overview of the orders / operations existing in the system.
- You wish to monitor the order progress for these orders / operations on the basis of the uploads recorded in the system.
- You wish to obtain an overview of the orders / operations completed in the previous shifts, the times required for this and the quantities produced.

Integration

The shop floor data / order data monitoring function uses order data that were either created directly in the system or that were transferred from other systems via interfaces.

Times and quantities entered in the BDE or MDE are accessed to display order progress.

Features

- Order overview:
 - Order overview as central overview function for the display of orders and their operations using a variety of selection criteria
 - Display of the order progress, i.e. all prepared, current, interrupted and completed OPs of a multi-level production order in the order overview
- Order information
 - Order information for the display of important information on an order, including its OPs, such as stock data, user fields, components, production resources and status information (degree of completion, remaining run time, quantities and times posted)
 - Display of comments entered at the terminal
- Operation overviews
 - Tabular list of the scheduled operations (status: prepared, interrupted)

- Tabular presentation of the currently logged on operations (status: running)
 - Tabular presentation of the operations already completed (status: completed)
 - Tabular presentation of all operations with predefined filter for display of logged on operations, completed operation and pool of orders
- Order shift log
 - Shift-related presentation of the completed operations
 - Graphic representation of quantities and times by article

2 Order information

Overview

HYDRA menu	Production control → Production overview → Order information
FEDRA menu	Detailed Scheduling → Current → Order information
Transaction code	orin
Function authorization	orin

Available user fields		
Where?	Object type/user field key	Source (type)
Table	AGNR/SYSTEM	Operation (MF-D)
Tab Order data → User fields	AUNR/depending on data record	order (MF-D)
Tab Operations → User fields	AGNR/depending on data record	Operation (MF-D)
Tab Operations → Components	MATLIST/SYSTEM	Material component (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

The *Order information* is the most important instrument for anyone in a position of responsibility in and around the production process. You use the *Order information* to monitor, control and oversee production.

Integration

The application *Order information* provides all data required to process operations in a clear and well-structured manner. The *Order information* therefore is one of the most important tools to control and monitor production. This application is very useful to prepare the work in production.

Selection criteria

The application provides the following selection criteria:

Order

This selection criterion refers to the order number. The selected order is displayed with its operations.

Show split OPs

The checkbox has an effect on the operations displayed. If the checkbox is enabled, the list of operations includes split operations.

In addition, the following information is shown for the order selected in the selection panel (view only):

Order status

Shows the status of the selected order in text form (highlighted in color)

Archived

If the order has already been archived, this is shown in text form behind the field displaying the order status.

Toolbar

When you call a function or target application, the parameters of the table are generally transferred. For this reason, you should always select an entry before calling an application.

Tab Main page



Update order (function authorization or.actualize)

Use the function [Update order](#) to update the order.



Schedule order (function authorization or.terminate)

Use the function [Schedule order](#) to schedule or reschedule the order.



Order overview (function authorization: orov)

Use this button to call the application [Order overview](#).



Documents

Use this function to show the documents stored for the operation.
Assign the documents in the application [Edit production resources and tools](#)



Serial numbers (function authorization: edser)

Opens the application [Edit serial numbers](#) for the order.



Inspection requirement (function authorization: irp)

Calls the application [Inspection requirement](#)



Inspection points (function authorization ipp)

Calls the application [Inspection points](#)

**Order network (function authorization ornet)**

Calls the application [Order network](#)

You can only open the applications *Inspection points*, *Inspection requirements* and *Order network*, if the extension [orin82](#) is activated.

Tab Edit**Edit orders (function authorization: edor*)**

Opens the application [Edit orders](#) for the order.

**Edit order sequences (function authorization: edseq)**

Opens the application [Edit order sequences](#) for the order.

**Edit operations (function authorization: edop)**

Opens the application [Edit operations](#) for the order.

**Edit long texts of operations (function authorization: edoptx)**

Opens the application [Edit long texts of operations](#) for the order.

**Edit components (function authorization: edopcomp)**

Opens the application [Edit components](#) for the order.

**Edit production resources and tools (function authorization: edopres)**

Opens the application [Edit production resources and tools](#) for the order.

Detail application *Order data*

The detail application *Order data* displays the order master data. Die different fields of the order header are described [here](#).

The sequence of the tabs can deviate from the one used in this document.

Detail application *Order data - Status*

The detail application *Order data - Status* shows the current status of the order.

- The times displayed are calculated using the sum total of all operations.
- The scrap quantity displayed is calculated using the sum total of all operations.

- The yield quantity displayed is the yield of the last operation that could be recorded.

For further information, see [here](#).

Detail application *Commentaries*

This detail application shows the BDE comments that have been added to the operations of the selected order during data collection (only applies when using HYDRA). This supports efforts towards a paperless production process.

The following data is available:

Recording time

Point in time (date, time) when the comment was recorded.

Collected at workplace

Workplace where the comment was recorded.

MES order number

The combined order and operation number of the operation the comment was recorded for.

Person

Personnel number of the person according to the HR master data who entered the comment.

Last name

Last name of the person according to the HR master data who entered the comment.

First name

First name of the person according to the HR master data who entered the comment.

Name

Entire name of the person (last name, middle name and first name) according to the HR master data who entered the comment.

Detail application *Operations*

The sequence of the tabs can deviate from the one used in this document.

Detail application *Operations* - *Operation*

The detail application *Operation* shows the operation's master data.

The fields of the operation are described [here](#).

Detail application *Operations* - *Operation status*

The detail application *Status* shows the current status of the relevant operation.

The operation's status information is described [here](#).

Detail application *Operations - Operation - RPA*

The detail application *Operation - RPA* shows the times recorded (actual values) for the relevant operation. The times are assigned to the different RPAs.

Detail application *Operations - Logged on to workplace*

The detail application *Logged on to workplace* shows the workplaces where the selected operation is logged on. If an operation is logged on to several workplaces at the same time, all these workplaces are shown here.

Workplace

Workplace

Workplace where the operation is logged on.

Login

Date

Time when the operation was logged on to the workplace.

Time

Time when the operation was logged on to the workplace.

Order

HYDRA order number

Combined order/operation number of the selected operation.

Detail application *Operations - Staff logged on*

This detail application shows the persons that are logged on to the operation selected above. Because several employees can be logged on to one operation, more than one person can be shown for an operation.

Person

Person

Personnel number of the person logged on to the operation (according to the HR master data). The field Workplace shows the workplace where the person is logged on with the specified operation.

Staff badge number

Staff badge number of the person logged on to the operation (according to the HR master data).

Last name

Last name of the person logged on to the operation (according to the HR master data).

First name

First name of the person logged on to the operation (according to the HR master data).

Name

Complete name of the person (last name, middle name and first name) logged on to the operation (according to the HR master data).

Login**Date**

Time when the operation was logged on to the workplace.

Time

Time when the operation was logged on to the workplace.

Advance logon

If the logon is currently still an advance logon recorded before the start of a shift, the flag in this column illustrates this.

Workplace**Workplace**

Workplace where the person is logged on with the currently selected operation.

Operator position/function

Operator function (identification) that the person uses to log on to the workplace (only relevant if a person logs on at the Windows terminal with an operator position).

Order**HYDRA order number**

Combined order/operation number of the selected operation.

Detail application *Operations - Resources logged on*

The detail application Resources logged on shows a table of tools and other resources needed for production that are directly or indirectly logged on to the operation.

Detail application *Operations - Material in use*

The detail application *Material in use* shows the input material logged on to the operation selected above.

Material

Batch number

Batch number of the material in use.

Material

Material number of the material in use according to the batch inventory.

Material designation/name

Material name of the material in use according to the batch inventory.

Material type

Material type of the material in use according to the batch inventory.

Quantities

Quantities

Original quantity of the material in use according to the batch inventory.

Remaining quantity

Remaining quantity of the material in use according to the batch inventory.

Unit

Quantity unit of the material in use according to the batch inventory.

Workplace

Workplace

Workplace where the input material/batch is logged on.

Login

Date

Time when the input material/batch was logged on to the workplace.

Time

Time when the input material/batch was logged on to the workplace.

Order

HYDRA order number

Combined order/operation number of the selected operation.

Detail application *Operations - Article in production*

This detail application shows the article (output batch) that is currently being produced with the operation.

Material

Batch number

HYDRA batch number of the article currently produced (output batch).

Material

Material number of the article currently produced (output batch) according to the batch inventory.

Material designation/name

Material name of the article currently produced (output batch) according to the batch inventory.

Material type

Material type of the article currently produced (output batch).

Quantities

Quantities

Quantity recorded so far and posted for the article (output batch).

Unit

Quantity unit of the article currently produced (output batch).

Workplace

Workplace

Workplace where the article (output batch) is logged on.

Login

Date

Time when the article (output batch) was logged on to the workplace.

Time

Time when the article (output batch) was logged on to the workplace.

Order

HYDRA order number

Combined order/operation number of the selected operation.

Detail application *Operations - PRT (production resources and tools)*

The detail application PRT shows all production resources and tools of an operation that have been transferred via interface. With operations that are not logged on, the detail application can also show the resources and tools that are not logged on to the operation.

Resource

Resource type

Resource type of the resource, referred to as production resources and tools

Resource

Identification

Designation

Name of the production resource according to the list of production resources and tools.

Comment 1, comment 2

Comment fields for the production resource according to the list of production resources and tools.

File

File name

If the production resource is a file (image, document), the file name is shown here.

Path

Logical reference to the path configuration

Quantities

Required quantity

Required quantity of the production resource or tool.

QU required quantity

Quantity unit of the required quantity of the production resource or tool.

Order

MES order number

Combined order/operation number of the selected operation.

Detail application Operations - Components

The detail application *Components* shows all consumption materials of an operation transferred via interface.

The fields of a component are described [here](#)

Detail application *Operations - Texts*

The detail application Texts shows additional texts for the operation selected.

Note:

The application only shows the text that fits the screen display (maximum).

No scroll function is provided for the text.

3 Order Overview

Overview

HYDRA menu	Production control → Production overview → Order overview
FEDRA menu	Detailed Scheduling → Current → Order overview
Transaction code	orov
Function authorization	orov

Available user fields		
Where	Object type/user field key	Source (type)
Table Order overview	AUNR/SYSTEM	Order (MF-D)
Table Order progress	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields?		Which user field types are available?

Purpose

You can use the *Order overview* function to get an overview of the production progress of an entire order integrating also the progress details of the different operations.

Integration

The workplace overview provides a production overview and is relevant to the production management. The order overview is the central dialog showing information about orders in the system. This dialog shows dates, quantities and times of production orders, overhead cost orders, maintenance orders and all other order types. The orders are displayed in a table. If you select a table row, the application shows the progress of the specified order.

Selection criteria

If you select more than one selection criteria, the order overview displays the data that matches all selection criteria.

The application provides the following selection criteria:

Order

This selection criterion refers to the order number. The application shows the selected order. You can also enter wildcards.

Order type

This selection criterion refers to the order type in the order header. The application only shows orders of the order type(s) selected.

Category

This selection criterion refers to the category of the order type in the order header. The application only shows the orders of the selected category(ies).

Final article

This selection criterion references the article in the order header. The application shows all orders that include the selected article. You can also use wildcards.

Article designation

This selection criterion refers to the article name (designation) defined in the order header. The application shows all orders assigned to the selected article name (designation). You can also use wildcards.

Order status

This selection criterion refers to the order statuses in the order header. The application only shows the orders assigned to the selected order status(es).

Control

This selection criterion refers to the control indicator in the order header. The application only shows the orders with the selected control indicator(s).

Sales order

This selection criterion relates to the sales order defined in the order header. The application shows all orders assigned to the selected sales order. You can also use wildcards.

Customer name

This selection criterion refers to the customer name (designation) defined in the order header. The application shows all orders with the selected customer name (designation). You can also use wildcards.

Project number

This selection criterion refers to the project number defined in the order header. The application shows all orders of the selected project number. You can also use wildcards.

Planned order

This selection criterion refers to the planned order defined in the order header. You can also use wildcards.

Cost object

This selection criterion refers to the cost object defined in the order header. The application shows all orders of the selected cost object. You can also use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. The application shows all orders of the selected MRP controller.

Order group

This selection criterion refers to the order group defined in the order header. The application shows all orders that are assigned to the selected order group.

Basic start date ... Basic start date

This selection criterion refers to the basic start date defined in the order header. The application only shows the orders, which are planned on or between the selected basic start dates.

Basic end date ... Basic end date

This selection criterion refers to the basic end date defined in the order header. The application only shows the orders, which are planned on or between the selected basic end dates.

Order end ... Order end

This selection criterion refers to the actual order end defined in the order header. The application only shows orders, which are planned on or between the selected order end dates.

Priority ... Priority

This selection criterion refers to the priority defined in the order header. The application shows all orders assigned to the selected priority.

Order index ... Order index

This selection criterion refers to the order index defined in the order header. The application shows all orders assigned to the selected order index.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the function authorization chkresp.

Order overview detail application**Status category****Order status**

The *status* column shows the bitmap ("LED") defined in the status configuration.

By default, the color of the status LED is the same as the color of the control LED.

Order status (text)

The current operation status specifies the status text.

Status since

Date/time since when the order is in this status.

Status since time

Date/time when the status was set.

Order start

Date/time of order start, i.e. date when the first operation of an order starts.

Order end

Date/time of order end, i.e. date when the last operation of an order was logged off.

Order category

Shows specific [Orders](#) data. Relevant fields are:

Order type

Displays the order type as text and symbol. The glossary describes standard order types. You can define additional order types when customizing the system.

Order

Number of the corresponding order.

User fields of the order header

Shows the configured user fields for the order overview in the *user fields* category.

Progress category**Number of OPs**

Shows the number of operations of the corresponding order.

Finished OPs

Shows the number of finished operations of the corresponding order.

Quantities category**Target quantity (B)**

Quantity specification for the order in base quantity unit.

Target scrap

Target scrap of the order in base quantity unit.

Unit

Defined unit (base quantity unit).

Yield (B)

Recorded yield in base quantity unit of the last operation that can be posted.



This is the last operation included in the order network. This operation is neither locked (internal control flag "Y") nor deleted logically (internal control flag "D"). This operation "provides" the quantity for the entire order.

This means that the yield is 0 as long as the order has not been finished, i.e. as long as you have not posted a quantity for the order's last operation.



If the last operation that allows quantity postings has a quantity > 0, and if the order overview does not show this quantity, then you must check the procedure described in the document [Activating_OrderRelatedKeyfigures.pdf](#) or proceed as described there.

Scrap (B)

Total of the scrap quantities entered for all operations of the order in base quantity unit.
Requirement: scrap is posted in base quantity unit for the operations.

Rework (B)

Total of the rework quantities entered for all operations of the order in base quantity unit.
Requirement: rework is posted in base quantity unit for the operations.

Open quantity (B)

Total of the open quantities entered for all operations of the order in base quantity unit.
Requirement: the open quantity is posted in base quantity unit for the operations.

Target times category**Planned lead time**

The planned lead time includes all planned execution times, like setup time, processing time, inspection time and retooling time (teardown).

Target setup time

Target setup time for the operation. The setup time is transferred by the ERP system or can be calculated using a formula that is defined during system customization.

Target processing time

Target processing time of the operation. The processing time is transferred by the ERP system or can be calculated using a formula defined during system customization.

Target execution time

Total of the target setup time + target processing time.

Target labor utilization

Total target labor utilization of all active OPs that can be posted.

Target wait time

Total wait time/idle time of all recordable, active OPs.

Transport time

Total transport time of all recordable, active OPs.

Waiting time

Total waiting time of all recordable, active OPs.

Delivery time

Total delivery time of all recordable, active OPs assigned to the "external processing" option.

Actual times category**Retention period of order**

The retention period of the order is calculated using the time between the first transfer of the order from the PPS system ("order release" = creation date of the order header in the system) and the actual logoff (time) of the last active operation of the order.

Note:

- The system cannot identify if the time of the order transfer is the time of the first transfer from the PPS system or if the order has been deleted and transferred a second time because of technical problems.
- If an order is transferred several times and the previous order is deleted with each transfer, the creation date of the order header shows the time of the last data transfer.

Lead Time

The lead time is the time between the first logon of an operation of the order and the logoff of the last active operation (time).

Setup time

The setup time of the order is the total of all setup times (RPA 7) of active operations where this data is collected.

Processing time

The processing time of the order is the total of the main production times (RPA 11) of all active operations where this data is collected.

Downtime

The order downtime is the total of the downtimes (RPA 1 to 6, RPA 8 to 10) of all active operations where this data is collected.

Occupancy time

The occupancy time is the total of the setup times (RPA 7), processing times (RPA 11) and downtimes (RPA 1...6, RPA 8...10) of all active operations where this data is collected.

Labor utilization

The labor utilization of the order is the total of personnel deployment times of all active operations where this data is collected.

Actual dates category**Order start date**

Date when the order was started, i.e. when the first operation of the order was logged on.

Order start time

Time when the order was started, i.e. when the first operation of the order was logged on.

Order end date

Date when the order was finished, i.e. when the last operation of the order was logged off.

Order end time

Time when the order was finished, i.e. when the last operation of the order was logged off.

Key figures category

Rate of capacity utilization (utilization efficiency)

The rate of capacity utilization is the ratio of the processing time (RPA 11) to the occupancy time (RPA 1 ... 11) in percent.

$$NGRAD = 100.0 * \frac{ANR.EGR : BMK11}{\sum_{n=1}^{11} ANR.EGR : BMKn}$$

Setup rate

The setup ratio is the ratio of the setup time (RPA 7) to the occupancy time (RPA 1...11) in percent.

Order progress detail application

The *order progress* detail application shows all operations of the above selected order.

If you select several orders in the order overview, this application shows all operations belonging to these selected orders. In this case, we recommend showing the "order" column.

Status category

Status

Shows the corresponding bitmap ("LED") defined in the status configuration.

By default, the color of the status LED is the same as the color of the control LED.

Status text

The current operation status specifies the status text.

Status since

Point in time since the status is available.

The field is empty with prepared operations.

Predecessor status

Status of the preceding operation. This status specifies whether the preceding operation has already been started. This is important if you want to know if material needed for the current operation has already been processed or produced.

Secondary status

Displays the currently set secondary status.

Please note: Use the [Configuration of the order status](#) to configure and use secondary statuses.

Order category**Order**

Shows the order number.

Operation category

Shows specific [Operations](#) data.

Note

Short text of the first note of an operation.

Primary quantity/secondary quantity/tertiary quantity category**Target quantity**

Quantity specifications for the [Operation](#).

Yield

The *yield* column shows the yield entered via the terminal or MOC.

Scrap

The *scrap* column shows the scrap entered via the terminal or MOC.

Rework

Quantity to be reworked.

Open quantity

The *open quantity* is another quantity account.

Unit

Quantity unit of the values displayed.



The quantities listed here are displayed as base, primary, secondary and tertiary quantities. In general, you should only show one of these quantity types. The terminal collects quantities in the *primary quantity*.

Postings category**First logon**

Date/time when the operation is logged on first.

Last interruption

Date/time when the operation was interrupted at last.

Last logoff

Date/time when the operation was logged off at last.

Last logon

Date/time when the operation was logged on at last

Specifications for production category**Planned for**

The *planned for* column shows whether the operation is planned

- for a *workplace* or
- if the operation is in the pool for the group (*planned for a group*).

Planned workplace

Workplace where the operation is planned. If the operation is planned for a *group*, this column indicates the workplace the operation was planned for most recently.

Group

Group where the operation is planned.

Tool, DNC, material

Main components assigned to the operation.

Dates category

Date specifications for the [Operation](#).

Miscellaneous category**Confirmation/upload number**

Confirmation (upload) number of the [Operation](#).

Target times category

Target time specifications for the [Operation](#).

Additional, calculated fields:

Total setup time

Total of the target setup time, additional setup time and target retooling (teardown) time.

Target execution time

Total of the target setup time (total) and the target processing time.

Actual times category**Setup time**

Posted setup time of the operation.

Actual execution time

Total of setup time plus processing time.

Processing time

Posted processing time (RPA 11) of the operation.

Downtime

Posted downtime of the operation

Total of downtimes (RPA 1..6, RPA 8..10)

Occupancy time

The occupancy time is the total of the setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1...6, RPA 8...10).

Lead Time

Posted occupancy time of the operation. The lead time is calculated using the time between the first logon of the operation and the last logoff of the operation.

Remaining run time

Shows the remaining production time. The system uses a formula and different parameters to calculate this value. The formula is stored with the operation. (RRT 1)

Remaining run time 2

Shows the remaining production time. The system uses a formula and different parameters to calculate this value. The formula is stored with the operation. (RRT 2)

Processing category

In addition to further target specifications for the [Operation](#), this category also provides the operation-related actual cycle.

Actual cycle

Calculate and display the actual cycle as follows:

- 1) OP prepared: by definition the actual cycle = 0.
- 2) OP logged on: the actual cycle corresponds to the current actual cycle from the Machine Data Collection (depending on the machine connection). The application [Workplaces/Machines](#) also shows this actual cycle.
- 3) OP interrupted or logged off/finished: the system calculates an average cycle time. This average cycle time is calculated by dividing the production time posted so far (RPA 11) by the strokes which have been recorded so far. The system only calculates the average cycle time for these order postings (no recalculation, e.g. after changes in the maintenance of postings dialog).

User fields of the order/operation

You can show the user fields of orders and operations in the table. The order overview can show orders or operations with different user field configurations. You must therefore use a defined [User field configuration](#) to create the columns. Create a category for the user fields in the table. All columns defined are first displayed in this category. You can change the place and visibility of these columns using the column configurator.

Toolbar

The function authorization required to execute the relevant function is entered in parentheses.

Order tab

**Order status (function authorization: or.statchg)**

Function to [change the order status](#).

**Update order (function authorization: or.actualize)**

Use the function [Update order](#) to update one or several selected orders.

**Schedule order (function authorization: or.terminate)**

Use the function [Schedule order](#) to schedule one or several selected orders.

**Order information (function authorization: orin)**

Calls the application [Order information](#) for the selected order.

**Edit orders (function authorization: edor)**

Calls the application [Edit orders](#) for the selected order.

**Inspection requirement (function authorization: irp)**

Click this button to call the application [Inspection requirement](#)

**Inspection points (function authorization ipp)**

Click this button to call the application [Inspection points](#)

**Order network (function authorization ornet)**

Click this button to call the application [Order network](#)

You can only open the applications *Inspection points*, *Inspection requirements* and *Order network*, if the extension [orov82](#) is activated.

Operation tab



Operation status (function authorization: op.statchg)

Function to [change the operation status](#).



Secondary status (function authorization: op.secstatchg)

Function to [change the secondary status of an operation](#).



Modify resource status (function authorization: op.resstatchg)

Function to [change the resource status of an operation](#)



Lock operation (function authorization: op.lock)

Use the button [Lock operation](#) to lock one or several selected operations.



Unlock operation (function authorization: op.unlock)

Use the button [Unlock operation](#) to unlock one or several selected operations.

Note

If you use the MPL module (Material and Production Logistics), you cannot use the below-mentioned functions to carry out batch-related order postings on the client because these postings are too complex.



Log on (function authorization: op.logon)

Use the [Log on](#) function to log on operations to the system.



Partial confirmation (function authorization: op.partconf)

Use the function [Post part quantities \(partial confirmation\)](#) to report part quantities for operations.



Interrupt (function authorization: op.interrupt)

Use the function [Interrupt](#) to interrupt operations.



Log off (function authorization: op.logoff)

Use the [Log off](#) function to log off operations.

**Terminate operation (function authorization: op.finish)**

Use the [Exit](#) function to log off interrupted or prepared operations.

**Reactivate (function authorization: op.reactivate)**

Use the function [Reactivate](#) to reactivate finished operations.

**Edit operations (function authorization: edop)**

Calls the application [Edit operations](#) for the selected order.



The functions "[Standard quantity for batches](#)" and "[Reaction to output batch change with quantity 0](#)" are not available as posting functions in the toolbar.

Person tab

Note

Use the operation-related functions *log on/interrupt/log off* for group workplaces.

**Log person on (function authorization: pn.logon)**

Use the function [Log person on](#) to log on an employee to an operation/machine.

**Log person off (function authorization: pn.logoff)**

Use the function [Log person off](#) to log off an employee from an operation/machine.

4 Order Shift Log

Overview

Menu	Order management → Production reports → Order shift log
Transaction code	ospr
Function authorization	ospr

Available user fields		
Where?	Object type/user field key	Source (type)
Table	AUNR/SYSTEM	Order (MF-D)
Table	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields?		Which user field types are available?

Purpose

The order shift log is a very useful tool for shift supervisors and foremen. It provides the information that persons responsible in production need for their daily reports.

The order shift log is a function of the production management. The individually configurable user interface presents shift-specific order data in a clear and comprehensive way.

The shift log evaluates the collected order data (quantities and times) in relation to the recorded shifts.

Integration

The order shift log shows all operations that meet the restrictions made in the selection panel. Only BDE log records of record type "U" and "E" are used.

This function provides the information about the operations completed during a shift. In addition to the quantities produced, the log also shows the times needed. The log provides this data for each order/operation.

You can correct the recorded order data using the function Order-related postings.

Selection criteria

The application provides the following selection criteria. If you request data, the system checks the responsibility area that includes the workplace of the relevant BDE log records.

Date from ... to ...

Enter a period of time to narrow down the log records displayed. The system selects the log records with a start date (logon date) in the period defined.

Default setting from "yesterday" until "today". The date is calculated based on the Gregorian calendar.

Shift: all, 1, 2, 3, 4

Within the entered period, only those log records are selected that are assigned to the shift entered according to the shift model.



At the time the data is selected, the system does not include operations that were logged on during the currently running shift, because no log records have been generated for them yet.

Workplace from ... to ...

Selects the log records that have been posted for the specified workplace. This workplace must be included in the responsibility area the user is authorized for. You can use wildcards in the field.

Group from ... to ...

Selects the log records that have been posted for the workplaces that are included in the user's responsibility area and that are assigned to the entered group. You can use wildcards in the field.

Cost center

Selects the log records that have been posted for the workplaces that are included in the user's responsibility area and that are assigned to the specified cost center. You can also use wildcards.

Report group

The selection criterion *Report group* refers to the report groups. The application shows all machines or workplaces that are assigned to the selected [Report group](#).

Order

Selects the log records that have been posted for the specified order.

Operation

Selects the log records that have been posted for the specified operation.

Article

Selects the log records that have been posted for operations with the specified article.

Tool

Selects the log records that have been posted for operations with the specified tool.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the function authorization chkresp.

Additional notes on the selection

Long-term data

If the selection period exceeds the period of time of the online data area, the system implicitly selects the data of the medium-term data area. You need not explicitly activate the access to the medium-term data area.

Order shift log detail application

The detail application provides the following fields:

Shift category

Shift date

Shift date of the shift that included the production of the operation (basis: BDE log record).

Shift

Shift in which the operation was produced (basis: BDE log record).

Order category

Category

Order category of the order/operation, e.g. production order (FA) or overhead cost order (GK).
(basis: backlog of orders).

Order type

Order type of the order (basis: backlog of orders).

Order

Order number of the order/operation (basis: backlog of orders).

Sequence

Sequence number of the operation (basis: backlog of orders, subject to customization/configuration).

Operation

Operation number of the operation (basis: backlog of orders).

Split

Split number of the operation, provided that it is a split operation (basis: backlog of orders, subject to configuration).

Operation designation

Designation of the operation (basis: backlog of orders).

Article/article designation

Article number and article designation of the operation (basis: backlog of orders).

Workplace category

Workplace/group/cost center

In addition to the order and article number, the workplace is displayed (including group and cost center of the workplace) where the operation has been produced in the selected shift. (Basis: workplace: BDE log record; group, cost center: workplace/resource master data).

Primary quantity category

Target quantity

Total target quantity of the operation in the relevant quantity unit (primary quantity unit, secondary quantity unit, tertiary quantity unit, base quantity unit – basis: backlog of orders).



For this column, no total is calculated. In some cases, one operation is produced during several shifts and here it is not correct to calculate totals.

Target quantity per shift

This field includes the theoretical target quantity for the posting. The duration of the posting may even be shorter than the shift duration if the operation, for example, was not logged on over the entire shift. If the posting covers the entire shift, the target quantity is calculated for each shift, otherwise the target quantity is only calculated for the shorter posting period.

$$SSKGUT = \frac{\sum_{01}^{11} ANR.EGR: BMK}{\left(\frac{ANR.SZY}{1000.0} \right)} * ANR.TLG$$

To calculate the values in the totals row, the system uses the totals of the RPA times and the average (not evaluated) of the target cycle values or the partitioning values (see totals row).

Yield

The yield posted for this operation – relating to the selected shift (basis: BDE log record).

Scrap

The scrap posted for this operation – relating to the selected shift (basis: BDE log record).

Rework

The rework quantity posted for this operation – relating to the selected shift (basis: BDE log record).

Open quantity

Open quantity posted for this operation – relating to the selected shift (basis: BDE log record).

Quantity unit

The respective quantity unit (basis: BDE log record).

Duration category

Target duration

The total operation-related target duration is calculated using the following formula (basis: backlog of orders):

$$\text{Solldauer} = \frac{\frac{\text{Sollzyklus [pro 1000]}_{AG}}{1000} * \text{Sollmenge (P)}_{AG}}{\frac{\text{Teiligkeit}_{AG}}{\text{Impulsfaktor}_{AG}}} + \text{statische Rüstzeit}_{AG}$$

If a pulse factor > 0 is not stored for the operation, the value 1 is preset.

If a partitioning > 0 is not stored for the operation, then the target duration is not calculated.



For this column, no total is displayed. In some cases, one operation requires several shifts for its production and a correct calculation of totals is then not possible.

Production

The production time recorded for this operation - relating to the selected shift (basis: BDE log record).

Downtime

The downtime recorded for this operation - relating to the selected shift (basis: BDE log record).

Total

Total of all production times and downtimes (sum of columns Production + Downtime).

RPA category

RPA

Detailed presentation of times recorded on the level of resource performance accounts . (Basis: BDE log record) .

Key figures category

Rate of capacity utilization

$$\text{NGRAD} = 100.0 * \frac{\text{ANR.EGR : BMK11}}{\sum_{01}^{11} \text{ANR.EGR : BMK}}$$

To calculate the values in the totals row, the system uses the totals of the RPA times (see totals row).

Output rate

$$\text{AUSBGD} = 100.0 * \left(\frac{\text{ANR.EGR : GUTP}}{1000.0 * \sum_{01}^{11} \text{ANR.EGR : BMK} * \text{ANR.TLG}} \right) * \text{ANR.SZY}$$

To calculate the values in the totals row, the system uses the totals of the RPA times and the yield quantities (P) as well as the arithmetic mean (unweighted) of the target cycle values or the partitioning values (see totals row).

Scrap rate

$$AQUOTE = 100.0 * \frac{ANR.EGR : A USP}{(ANR.EGR : GUTP + ANR.EGR : A USP)}$$

To calculate the values in the totals row, the system uses the totals of the yield quantities (P) and the scrap quantities (P) (see totals row).

Assignment utilization rate

$$BNGRAD = 100.0 * \frac{ANR.EGR : BMK11}{\left(\begin{array}{l} ANR.EGR : BMK01 + ANR.EGR : BMK02 + ANR.EGR : BMK03 \\ + ANR.EGR : BMK04 + ANR.EGR : BMK05 + ANR.EGR : BMK08 \\ + ANR.EGR : BMK09 + ANR.EGR : BMK10 + ANR.EGR : BMK11 \end{array} \right)}$$

To calculate the values in the totals row, the system uses the totals of the RPA times (see totals row).



The KPIs above can be changed according to the customer's requirements using the formula management. For the KPI definition, the acronyms below are available.

Acronyms for the formula definition	Description
ANR.EGR:BMK01 ANR.EGR:BMK02 ANR.EGR:BMK03 ANR.EGR:BMK04 ANR.EGR:BMK05 ANR.EGR:BMK06 ANR.EGR:BMK07 ANR.EGR:BMK08 ANR.EGR:BMK09 ANR.EGR:BMK10 ANR.EGR:BMK11 ANR.EGR:BMK12	Times recorded for the resource performance accounts 1-12
ANR.EGR:AUSP ANR.EGR:AUSB ANR.EGR:AUST ANR.EGR:AUSS	Scrap quantity recorded in the units primary, secondary, base and tertiary
ANR.EGR:GUTP ANR.EGR:GUTB ANR.EGR:GUTT ANR.EGR:GUTS	Yield quantity recorded in the units primary, secondary, base and tertiary
ANR.EGR:PRBP ANR.EGR:PRBB ANR.EGR:PRBT ANR.EGR:PRBS	Open quantity recorded in the units primary, secondary, base and tertiary
ANR.EGR:NCHP ANR.EGR:NCHB ANR.EGR:NCHT ANR.EGR:NCHS	Rework quantity recorded in the units primary, secondary, base and tertiary

ANR.EGR:HUBG ANR.EGR:HUB	Cycles recorded and cycles recorded for yield.
ANR.SZY ANR.TLG ANR.IMPFAKT	Target cycle, partitioning and pulse factor of the operation.

Machine category

Target cycle

Time planned for 1000 cycles of the machine. This value is the default value of the HYDRA-MDE machine monitoring (cycle monitoring). The value specified for the operation is used to populate this field. If the target cycle is changed when the operation is logged on, the change is not integrated.

For this column, the totals row shows the arithmetic mean (unweighted) of all target cycle values. This value is also used in the totals rows of the different key figures.

Actual cycle

$$IZYCLC = 1000.0 * \frac{ANR.EGR : BMK11}{ANR.EGR : HUBG}$$

For this column, the totals row shows the arithmetic mean (unweighted) of all actual cycle values.



Note: The duration in the field RPA 11 depends on the workplace setting *Posting of machine time with operations logged on simultaneously*. If a proportional posting is configured here, also the duration shown in this field is a proportional duration.

Partitioning

Integer value that specifies the number of parts produced per clock pulse. The value specified for the operation is used to populate this field. If the partitioning is changed when the operation is logged on, the change is not integrated.

For this column, the totals row shows the arithmetic mean (unweighted) of all partitioning values. This value is also used in the totals rows of the different key figures.

Total cycles

Number of cycles recorded while the operation was logged on. For this column, no total is calculated.

Pulse factor

Integer value that specifies the number of parts produced per clock pulse. The value specified for the operation is used to populate this field. For this column, no total is calculated.

Note

Displaying the actual quantities and the actual durations (RPA)

The shift automatic option is not available for workplaces that are not assigned to any terminal or that are assigned to a terminal configured as a BDE terminal (shop floor data collection). This means that there are no automatic order or person-related postings at the end of shifts. In this case, you cannot exactly assign the recorded quantities and durations to the shifts. The system therefore assigns quantities and durations proportionally. This assignment is based on the workplace's shift calendar.

Example:

In the example, the shift model is as follows: shift 1: 6:00 to 14:00 ; and shift 2: 14:00 to 22:00. This shift model is assigned to a workplace matching the above-mentioned criteria. An OP has been logged on at 13:00 and off at 16:00. For the OP logoff, 90 is uploaded as the yield.

In this case, the shift log for the operation in shift 1 will calculate an order duration of 60 minutes and a yield of 30. For shift 2, an order duration of 120 minutes is calculated and a yield of 60. The RPA-related durations are also calculated based on the shift model that the workplace is based on.



At the time the data is selected, the system does not include operations that were logged on during the currently running shift, because no log records have been generated for them yet.

Durations acc. to article detail application

The detail application *Durations acc. to article* shows the durations that have been posted for the article in a bar chart. Only the operations selected in the detail application *Order shift log* are used to calculate the values.

The bar chart shows the article numbers of the selected operations on the y-axis and absolute values (durations) are displayed on the x-axis. The respective quantity accounts specify the color of the bars (production/RPA 11: green; downtimes/RPA 1-11: red). Bars are sorted in descending order by production duration.

You can use a multi-combo box to define the durations that are shown as a bar:

- Production
- Downtimes

The bars are shown in a "stacked" form so that the total quantity can be defined differently for each user.

Activate the check box "Show labels", to show the values on the bars. Note: These labels are displayed for the selected duration.

Quantities acc. to article detail application

The detail application *Quantities acc to article* shows the quantities that have been posted for the article in a bar chart. Only the operations selected in the detail application *Order shift log* are used to calculate the values.

The bar chart shows the article numbers of the selected operations on the y-axis and absolute values (quantities) are displayed on the x-axis. Bars are sorted in descending order by production duration.

Toolbar

When you call a function or target application, the parameters of the table are generally transferred. For this reason, always select an entry before calling an application.



Order information (function authorization: orin)

Use this button to call the application [Order information](#).



Order overview (function authorization: orov)

Use this button to call the application [Order overview](#).

5 Pool of Orders

Overview

HYDRA menu	Production control → Production support → Pool of orders
FEDRA menu	Detailed Scheduling → Current → Pool of orders
Transaction code	plop
Function authorization	plop

Available user fields		
Where	Object type/user field key	Source (type)
Table	AUNR/SYSTEM	Order (MF-D)
Table	AGNR/SYSTEM	Operation (MF-D)

Purpose

The *Pool of orders* application is the most important dialog for foremen and supervisors.

The *Pool of orders* provides a selection of operations that is clearly sorted by categories. These operations are included in the pool of a machine or group.

Integration

The pool of orders shows all operations that match the specified selections.

Generally, only unplanned and planned operations are displayed in the pool of orders. The system identifies these operations via the control indicators V, U or S. The operations usually have the operation status "prepared", "interrupted" or "not free" (depending on the project).

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion specifies the workplace stored for the operation. All operations assigned to the selected workplaces are displayed. You can also use wildcards.

Group from ... to ...

This selection criterion refers to the group that is defined for the operation. The system displays all operations that are assigned to the selected group. It does not matter if the operations are already scheduled for a workplace or not. You can also use wildcards.

Planned for

You can use this option to specify if only operations are selected that have been scheduled for a workplace or operations that are still in the pool of groups.

The column "planned for" shows whether the operation is planned

- for a group (pool of groups) or
- for a workplace (pool of workplaces).

Order

This selection criterion refers to the order number. Only operations with the specified order number are displayed. You can also use wildcards.

Article

This selection criterion refers to the article of the operation. The application displays all operations having assigned the specified article. You can also use wildcards.

Article name/designation

This selection criterion refers to the article of the operation. The application displays all operations that match the specified article designation. You can also use wildcards.

Operation status

Operation status.

Notes:

- The selection list shows ALL configured workplace statuses. Also statuses that are not relevant for this application are displayed.
- The system can only display the operations with the selected operation status that are also included in the selection of the control.

Predecessor status

Status of the preceding operation.

Control

Control indicator of the current operation status.

Notes:

- The selection list shows ALL configured control indicators. Also control indicators that are not relevant for this application are displayed.
- By default, the values Prepared, Interrupted and None (short V, U, S) are preassigned in the selection panel.

Category

This selection criterion refers to the category of the operation's order type. The application only displays operations that match the specified category of order types.

Order type

This selection criterion refers to the order type of the operation. Only operations assigned to the selected order type are displayed. The selection list displays ALL configured order types with the respective order type name.

Processing code

This selection criterion refers to the processing code of the operation. Only operations with the selected processing code are displayed. You can also use wildcards.

Priority

This selection criterion refers to the priority of the operation. Only operations with the selected priority are displayed.

OP

This selection criterion refers to the operation number. Only operations with the specified operation number are displayed. You can also use wildcards.

Tool

This selection criterion refers to the tool used for the operation. All operations with the specified tool are displayed. You can also use wildcards.

Show split OPs

If this option is enabled, split operations are additionally displayed.

Customer name

This selection criterion refers to the customer designation stored for the operation. The application shows all orders with the selected customer name (designation).

Cost center

This selection criterion refers to the cost center of the workplace defined for the operation. The system displays all operations assigned to workplaces of the selected cost center. You can also use wildcards.

Planned start ... to ...

This selection criterion refers to the planned start date defined for the operation. The system only displays operations with a planned start on or between the specified date(s).

Scheduled start time ... to ...

This selection criterion refers to the scheduled start date of the operation. The system only displays operations with a scheduled start date on or between the specified date(s).

Earliest start ... to ...

This selection criterion refers to the earliest start date of the operation. The system only displays operations with an earliest start on or between the specified date (s).

Latest end ... to ...

This selection criterion refers to the latest end date of the operation. The system only displays operations with a latest end on or between the specified date(s).

Actual start ... to ...

This selection criterion refers to the start date of the operation. The system only displays operations with a start date on or between the specified date(s).

Actual end ... to ...

This selection criterion refers to the end date of the operation. The system only displays operations with an end date on or between the specified date(s).



Use the MOC application "[Order overview](#)" to select the operations using the actual dates of the order – order start or order end.

Basic date start ... to ...

This selection criterion refers to the basic start date of the order. The system only displays operations with a basic start date on or between the specified date(s).

Basic date end ... to ...

This selection criterion refers to the basic end date of the order. The system only displays operations with a basic end date on or between the specified date(s).

Sales order

This selection criterion refers to the sales order defined in the order header. All operations assigned to the selected sales order are displayed.

Order index ... to ...

This selection criterion refers to the order index defined in the order header. The system displays all operations with the specified order index in the order header.

Planned order

This selection criterion refers to the planned order number from SAP that is defined in the order header.

Project number

This selection criterion refers to the project number of the operation. The system displays all orders with the specified project number.

Order group

This selection criterion refers to the order group of the order header. The system displays all operations with the specified order index in the order header. You can also use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. The system displays all operations with the specified MRP controller in the order header. You can also use wildcards.

MOP

Restriction to a merged operation.
You can also use wildcards.

Show MOP

Defines the [Merged operations](#) to be displayed.
By default, the system displays merged operations and individual operations; operations that are summarized in a MOP are not displayed.

Check responsibility area

The user can only use this option if the respective license and function authorization "filterProdInd" is available and if the function authorization "chkresp" is enabled. With this selection option, the user can specify if data is displayed using the responsibility area of the workplace or of the object operation/order.

"Pool of orders" detail application

The detail application provides the following fields:

Status category**Status**

Shows the respective bitmap ("LED") defined in the status configuration.
The totals line displays the number of operations in this column.

Status text

The current operation status specifies the status text.

Status since

Specifies date and time when the respective status has been set.

Predecessor status

Status of the preceding operation. This status specifies whether the preceding operation has already been started. This is important if you want to know if material has already been processed or produced that you need for the current operation.

Secondary status

Displays the currently set secondary status.



The use of the secondary status requires a respective [Configuration](#).

"Order" category

This category displays specific data for [operations](#) and [orders](#). Relevant fields are:

Order type

Displays the order type as text and symbol.

The order types of the HYDRA standard are described in the glossary. You can [configure](#) further order types.

Order

Specifies the number of the respective order.

Sequence

Order sequence (only relevant if sequences are used)

OP

Specifies the number of the respective operation

Split

Specifies the split number if the operation is a split operation.

Current number of splits

Number of splits of an operation if the operation has been split.

Note

Short text of the first note of an operation.

Printed

This column shows whether the respective time ticket has already been printed.

Specifications for production category

The relevant fields of this category are as follows:

Planned workplace

Workplace where the operation is planned.

Planned for

The "planned" column shows whether the operation is planned

G for a group (pool of groups) or

M for a workplace (pool of workplaces).

Group

Group for which the operation is planned.

"OP dates" category

This category provides date specifications for the [operation](#).

"Remaining run time" category**Remaining run time**

This column specifies the remaining production time. The system calculates this value using a formula based on different parameters. The formula used is stored for the operation.

"Target times" category

This category displays the target time specifications for the [operation](#)

Additional calculated fields:**Total setup time**

Sum total of setup time, additional setup time and retooling/teardown time.

Target execution time

Sum total of setup time (total) plus processing time.

Primary quantity/secondary quantity/tertiary quantity category**Target quantity**

Quantity specifications for the [operation](#).

Yield

Yield posted for the operation.

Scrap

Scrap posted for the operation.

Rework

Quantity posted for the operation that must be reworked.

Problem quantity

Problem quantity posted for the operation.

Unit

Quantity unit of the values displayed.



The quantities listed here are displayed as base, primary, secondary and tertiary quantities. It is usually best to show only one of the quantity types. The terminal collects quantities in *primary quantity*.

Toolbar



Generate merged operation (function authorization: op.colopcreate)

Function [Generate merged operation](#)

Selected operations are combined to form a merged operation. Instead of the different single operations, the merged operation is logged on to the terminal and displayed on the terminal.



Cancel merged operation (function authorization: op.coloprelease)

Function [Cancel merged operation](#)

The merged operation is undone and the separate operations are generated.



Order information (function authorization: orin)

This button calls the application [Order information](#).



Order overview (function authorization: orov)

This button calls the application [Order overview](#).



Schedule controlling: save baseline plan (function authorization: esvb)

Function [Schedule controlling: save baseline plan](#)



Shop floor paper (function authorization: repsfpap)

Function [Print shop floor papers](#)



Time ticket (function authorization: reptimet)

Function [Print time tickets](#)

6 Operations Logged on

Overview

HYDRA menu	Production control → Production overview → Running operations (operations logged on)
FEDRA menu	Detailed Scheduling → Current → Operations logged on
Transaction code	rop
Function authorization	rop

Available user fields		
Where	Object type/user field key	Source (type)
Table	AUNR/SYSTEM	Order (MF-D)
Table	AGNR/SYSTEM	Operation (MF-D)

Purpose

The function *Operations logged on* shows the currently running and automatically interrupted (by change of shifts) operations (with the control indicators L, F).



Please note that you have to set the option "Show the operations logged on for each workplace" if you want the application to show the logged on operations (status "running") for each workplace.

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace stored in the operation. The application shows all operations assigned to the selected workplaces. You can also use wildcards.



For operations that are logged on, this does not refer to the workplace where the operation is currently logged on. The application shows the workplace where the operation is planned.

Group from ... to ...

This selection criterion refers to the group that is defined for the operation. The application shows all operations assigned to the selected group. You can also use wildcards.



For operations that are logged on, this does not refer to the group of the workplace where the operation is currently logged on. The application shows the group where the operation is planned.

Planned for

Use this option to only show operations that have initially been planned for a workplace or a group.

Order

This selection criterion refers to the order number. The application only shows operations with the specified order number. You can also use wildcards.

Article

This selection criterion refers to the article of the operation. The application displays all operations that are assigned to the specified article. You can also use wildcards.

Article designation

This selection criterion refers to the article name (designation) defined in the operation. The application displays all operations that match the specified article name. You can also use wildcard characters.

Operation status

Current operation status.

Note:

The selection list shows ALL configured operation statuses; even those statuses that are not relevant for this application.

Predecessor status

Status of the preceding operation.

Control

Current control indicator of the operation.

Note:

The selection list shows ALL configured control indicators; even those indicators that are not relevant for this application.

Category

This selection criterion refers to the category of the operation's order type. The application only displays operations with an order type of the specified category.

Order type

This selection criterion refers to the order type of the operation. The application only shows operations with the selected order type.

Processing code

This selection criterion refers to the processing code of the operation. The application only shows operations with the selected processing code. You can also use wildcards.

Priority

This selection criterion refers to the priority of the operation. The application only shows operations with the selected priority.

OP

This selection criterion refers to the operation number. The application only shows the operations with the specified operation number. You can also use wildcards.

Tool

This selection criterion refers to the tool defined for the operation. The application shows all operations that are assigned to the specified tool. You can also use wildcards.

Show split OPs

If you enable this checkbox, the application also shows split operations.

Customer name

This selection criterion refers to the customer name (designation) defined in the operation. The application shows all orders with the selected customer name (designation).

Cost center

This selection criterion refers to the cost center of the workplace defined for the operation. The application shows all operations assigned to the selected cost center. You can also use wildcards.

Planned start ... to ...

This selection criterion refers to the planned start date defined for the operation. The application only shows the operations whose planned start date is included in the selected period.

Scheduled start time ... to ...

This selection criterion refers to the scheduled start date of the operation. The application only shows the operations whose scheduled start date is included in the selected period.

Earliest start ... to ...

This selection criterion refers to the earliest start date of the operation. The application only shows the operations whose earliest start date is included in the selected period.

Latest end ... to ...

This selection criterion refers to the latest end date of the operation. The system only displays the operations whose latest end date is included in the selected period.

Actual start ... to ...

This selection criterion refers to the start date of the operation. The system only displays the operations whose start date is included in the selected period.

Actual end ... to ...

This selection criterion refers to the end date of the operation. The system only displays the operations whose end date is included in the selected period.



Use the MOC application "[Order overview](#)" to select operations using the actual dates of the order, i.e. the order start or order end.

Basic date start ... to ...

This selection criterion refers to the basic start date of the order. The application only displays the operations whose basic start date of the order is included in the selected period.

Basic date end ... to ...

This selection criterion refers to the basic end date of the order. The application only displays the operations whose basic end date of the order is included in the selected period.

Sales order

This selection criterion refers to the sales order defined in the order header. The application shows all operations that are assigned to the selected sales order.

Order index ... to ...

This selection criterion refers to the order index defined in the order header. The application displays all operations whose order index of the order header matches the selected order index.

Planned order

This selection criterion refers to the planned SAP order number that is defined in the order header.

Project number

This selection criterion refers to the project number of the operation. The application displays all orders that are assigned to the selected project number.

Order group

This selection criterion refers to the order group of the order header. The application displays all operations whose order group of the order header matches the selected order group. You can also use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. The application shows all operations whose MRP controller of the order header matches the entered MRP controller. You can also use wildcards.

MOP

Use this option to only show merged operations.
You can also use wildcards.

Show MOP

Use this option to specify the [merged operations](#) to be displayed.

By default, the application shows merged operations and individual operations. The application does not show the operations that are summarized in an MOP.

Show the operations logged on for each workplace

If you check this option, the application shows an operation multiple times if this OP is simultaneously logged on to multiple workplaces.

The category "logged on to workplace" (hidden by default) shows information about the workplace where the operation is logged on.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. To use this selection option, you require the function authorization chkresp.

Operations logged on detail application

The table shows the below-mentioned fields, among other things. For further information refer to the following documents

- [Existing order data](#)
- [Current data and KPIs of the operation](#)

Status category**Status**

The *status* column shows the bitmap ("LED") defined in the status configuration.

The totals line of this column displays the number of operations.

Status text

The current operation status specifies the status text.

Status since

Specifies the date and time since when the status has been set.

Predecessor status

Status of the preceding operation. This status specifies whether the preceding operation has already been started. This is important if you want to know if material needed for the current operation has already been processed or produced.

Secondary status

Displays the currently set secondary status.



Optionally, you can configure and use secondary statuses while customizing HYDRA.

Order category

Displays specific data for the [Operations](#) and [Orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol.

The glossary describes the standard order types. Further order types can be defined as part of a system customization.

Category

The category summarizes similar order types. The following categories are available in the system:

FA = production order

PJ = project order

PM = maintenance order

KP = capacity order

GK = overhead order

Order

Shows the order number.

Sequence

Order sequence (only relevant if sequences are used).

OP

Shows the operation number.

Split

Shows the split number if the operation is a split operation.

Current number of splits

Shows the number of splits of an operation if the operation has been split.

Note

Shows the short text of the first note of an operation.

Printed

This column shows whether the time ticket has already been printed.

Specifications for production category

The relevant fields are:

Planned workplace

Workplace where the operation is planned.

Planned for

The "planned for" column shows whether the operation is planned

G for a group (pool of groups) or

M for a workplace (pool of workplaces).

Group

Group where the operation is planned.

OP dates category

This category provides date specifications for the [Operation](#).

Remaining run time category**remaining run time**

Shows the remaining production time. The system uses a formula and different parameters to calculate this value. The formula is stored with the operation.

Target times category

This category displays the target time specifications for the [Operation](#).

Additional, calculated fields:**Total setup time**

Sum total of setup time, additional setup time and retooling/teardown time.

Target execution time

Total of setup time plus processing time.

Primary quantity/secondary quantity/tertiary quantity category**Target quantity**

Quantity specifications for the [Operation](#).

Yield

The *yield* column shows the yield entered via the terminal or MOC.

Scrap

The *scrap* column shows the scrap entered via the terminal or MOC.

Rework

Quantity to be reworked.

Problem quantity

The problem quantity is another quantity account.

Unit

Quantity unit of the values displayed.



The quantities listed here are displayed as base, primary, secondary and tertiary quantities. In general, you should only show one of these quantity types. The terminal collects quantities in the *primary quantity*.

Postings category**First logon**

Shows the date and time when the operation was logged on for the first time.

Last logoff

Shows the date and time when the operation was logged off the last time. This kind of information is useful if an operation was reactivated.

Date of last posting

Date of the last posting made for this operation.

Time of last posting

Time of the last posting made for this operation.

Logged on to workplace category

The below-mentioned fields pertaining to this category are only completed, if you set the option "Show the operations logged on for each workplace".

Workplace

Workplace where the operation is currently logged on.

Short name

Short name of the workplace.

Name

Name/description of the workplace.

Group

Group of the workplace.

Cost center

Cost center of the workplace.

Toolbar

When you call a function or target application, the parameters of the table are always transferred. For this reason, always select an entry to call an application.



Order information (function authorization: orin)

Use this button to call the application [Order information](#).



Order overview (function authorization: orov)

Use this button to call the application [Order overview](#).

7 Finished Operations

Overview

HYDRA menu	Order Management → Production reports → Finished operations
FEDRA menu	Detailed Scheduling → Evaluations → Finished operations
Transaction code	fop
Function authorization	fop

Available user fields		
Where?	Object type/user field key	Source (type)
Table	AUNR/SYSTEM	Order (MF-D)
Table	AGNR/SYSTEM	Operation (MF-D)

Purpose

The application *Finished operations* provides a clearly categorized selection of finished operations to the shift manager, supervisor or foreman.

Integration

The application shows the operations that have been selected in the selection panel.

All operations are displayed that have the control indicators "E" (finished), "A" (archived) and "D" (logically deleted).

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace stored in the operation. The application shows all operations assigned to the selected workplaces. You can also use wildcards.

Group from ... to ...

This selection criterion refers to the group that is defined for the operation. The application shows all operations assigned to the selected group. You can also use wildcards.

Planned for

Use this option to only show operations that have initially been planned for a workplace or a group.

Order

This selection criterion refers to the order number. The application only shows operations with the specified order number. You can also use wildcards.

Article

This selection criterion refers to the article of the operation. The application displays all operations that are assigned to the specified article. You can also use wildcards.

Article designation

This selection criterion refers to the article of the operation. The application displays all operations that match the specified article name. You can also use wildcard characters.

Operation status

Current operation status.

Note:

The selection list shows ALL configured operation statuses; even those statuses that are not relevant for this application.

Predecessor status

Status of the preceding operation.

Control

Current control indicator of the operation.

Note:

The selection list shows ALL configured control indicators; even those indicators that are not relevant for this application.

Category

This selection criterion refers to the category of the operation's order type. The application only displays operations with an order type of the specified category.

Order type

This selection criterion refers to the order type of the operation. The application only shows operations with the selected order type.

Processing code

This selection criterion refers to the processing code of the operation. The application only shows operations with the selected processing code. You can also use wildcards.

Priority

This selection criterion refers to the priority of the operation. The application only shows operations with the selected priority.

OP

This selection criterion refers to the operation number. The application only shows the operations with the specified operation number. You can also use wildcards.

Tool

This selection criterion refers to the tool defined for the operation. The application shows all operations that are assigned to the specified tool. You can also use wildcards.

Show split OPs

If you enable this checkbox, the application also shows split operations.

Customer name/designation

This selection criterion refers to the customer name stored in the operation. The application shows all orders with the selected customer name (designation).

Cost center

This selection criterion refers to the cost center of the workplace stored in the operation. All operations assigned to the selected cost center are displayed. You can also use wildcards.

Planned start ... to ...

This selection criterion refers to the planned start date defined for the operation. The application only shows the operations whose planned start date is included in the selected period.

Scheduled start time ... to ...

This selection criterion refers to the scheduled start date of the operation. The application only shows the operations whose scheduled start date is included in the selected period.

Earliest start ... to ...

This selection criterion refers to the earliest start date of the operation. The application only shows the operations whose earliest start date is included in the selected period.

Latest end ... to ...

This selection criterion refers to the latest end date of the operation. The system only displays the operations whose latest end date is included in the selected period.

Actual start ... to ...

This selection criterion refers to the start date of the operation. The system only displays the operations whose start date is included in the selected period.

Actual end ... to ...

This selection criterion refers to the end date of the operation. The system only displays the operations whose end date is included in the selected period.



Use the MOC application "[Order overview](#)" to select operations using the actual dates of the order, i.e. the order start or order end.

Basic date start ... to ...

This selection criterion refers to the basic start date of the order. The application only displays the operations whose basic start date of the order is included in the selected period.

Basic date end ... to ...

This selection criterion refers to the basic end date of the order. The application only displays the operations whose basic end date of the order is included in the selected period.

Sales order

This selection criterion refers to the sales order defined in the order header. The application shows all operations that are assigned to the selected sales order.

Order index ... to ...

This selection criterion refers to the order index defined in the order header. The application displays all operations whose order index of the order header matches the selected order index.

Planned order

This selection criterion refers to the planned SAP order number that is defined in the order header.

Project number

This selection criterion refers to the project number of the operation. The application displays all orders that are assigned to the selected project number.

Order group

This selection criterion refers to the order group of the order header. The application displays all operations whose order group of the order header matches the selected order group. You can also use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. The application shows all operations whose MRP controller of the order header matches the entered MRP controller. You can also use wildcards.

MOP

Use this option to only show merged operations. You can also use wildcards.

Show MOP

Use this option to specify the [merged operations](#) to be displayed.

By default, the application shows merged operations and individual operations. The application does not show the operations that are summarized in an MOP.

Check responsibility area

The user can only use this option if the respective license and function authorization "filterProdInd" is available and if the function authorization "chkresp" is enabled. Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data.

Detail application: Finished Operations

The detail application provides the following fields:

Status category**Status**

The *status* column shows the bitmap ("LED") defined in the status configuration.

The totals line of this column displays the number of operations.

Status text

The current operation status specifies the status text.

Status since

Specifies the date and time since when the status has been set.

Predecessor status

Status of the preceding operation. This status specifies whether the preceding operation has already been started. This is important if you want to know if material needed for the current operation has already been processed or produced.

Secondary status

Displays the currently set secondary status.



Optionally, you can configure and use secondary statuses while customizing the system.

If an X is displayed here it is an operation of an order that has been completed technically in SAP.

Order category

Displays specific data for the [operations](#) and [orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol.

The glossary describes the standard order types. You can define further order types as part of a customization.

Order

Shows the order number.

Sequence

Order sequence (only relevant if sequences are used).

OP

Shows the operation number.

Split

Shows the split number if the operation is a split operation.

Current number of splits

Shows the number of splits of an operation if the operation has been split.

Note

Shows the short text of the first note of an operation.

Printed

This column shows whether the time ticket has already been printed.

Specifications for production category

Specifications for production with respect to the machine, tool, DNC, material for the [Operation](#).

Target times category

This category displays the target time specifications for the [Operation](#).

Additional, calculated fields:**Total setup time**

Sum total of setup time, additional setup time and retooling/teardown time.

Target execution time

Total of setup time plus processing time.

Processing category

Default target specifications for the production of the [Operation](#)

Primary quantity/secondary quantity/tertiary quantity category**Target quantity**

Quantity specifications for the [Operation](#).

Yield

The *yield* column shows the yield entered via the terminal or client.

Scrap

The *scrap* column shows the scrap entered via the terminal or client.

Rework

Quantity to be reworked.

Problem quantity

The problem quantity is another quantity account.

Unit

Quantity unit of the values displayed.



The quantities listed here are displayed as base, primary, secondary and tertiary quantities. In general, you should only show one of these quantity types. The terminal collects quantities in the *primary quantity*.

OP dates category

This category provides date specifications for the [Operation](#).

Actual times category

[Actual times](#) currently collected for the operation.

Key performance indicators category

The formulas used to calculate the values are described [here](#).

Lock category**Lock**

This option specifies if the operation is locked.

If an operation is locked, you cannot log this operation on and it is not displayed in the sequencing list on the terminal.

Locked by and on

Specifies the time and the user who last locked the operation.

Unlocked by and on

Specifies the time and the user who last unlocked the operation.

Toolbar

When you call a function or target application, the parameters of the table are always transferred. For this reason, always select an entry to call an application.

**Order information (function authorization: orin)**

Use this button to call the application [Order information](#).

**Order overview (function authorization: orov)**

Use this button to call the application [Order overview](#).

8 Operations

Overview

HYDRA menu	Production control → Production overview → Operations
FEDRA menu	Detailed Scheduling → Current → Operations
Transaction code	op
Function authorization	aop

Available user fields		
Where	Object type/user field key	Source (type)
Table	AUNR/SYSTEM	Order (MF-D)
Table	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

The *Operations* application is an important dialog for shift supervisors, assistant foremen or foremen. This application shows operations grouped by their processing status in a clearly structured manner.



Please note that you have to set the option "Show the operations logged on for each workplace" if you want the application to show the logged on operations (status "running") for each workplace.

Integration

This application provides the user with a clearly categorized selection of operations. The configurable client user interface enables the user to view the information at large, in detail or clearly structured at a glance.

Note: You can use the application [Order overview](#) to identify the progress of an order, i.e. all operations pertaining to an order are displayed.

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace planned for the operation. The application shows all operations that are assigned to the selected workplace.

You can also use wildcards.



For operations that are logged on, this does not refer to the workplace where the operation is currently logged on. The application shows the workplace where the operation is planned.

Group from ... to ...

This selection criterion refers to the group that is planned for the operation. The application shows all operations that are assigned to the selected group.

You can also use wildcards.



For operations that are logged on, this does not refer to the group of the workplace where the operation is currently logged on. The application shows the group where the operation is planned.

Planned for

Use this option to only show operations that have initially been planned for a workplace or a group.

Order

This selection criterion refers to the order number. The application only shows operations with the specified order number.

You can also use wildcards.

Article

This selection criterion refers to the article of the operation. The application displays all operations that are assigned to the specified article.

You can also use wildcards.

Article designation

This selection criterion refers to the article name (designation) defined in the operation. The application displays all operations that match the specified article name. You can also use wildcard characters.

Operation status

Current operation status.

Predecessor status

Status of the preceding operation.

Control

Current control indicator of the operation

Category

This selection criterion refers to the category of the operation's order type. The application only displays operations with an order type of the specified category.

Order type

This selection criterion refers to the order type of the operation. The application only shows operations with the selected order type.

Processing code

This selection criterion refers to the processing code of the operation. The application only shows operations with the selected processing code.

You can also use wildcards.

Priority

This selection criterion refers to the priority of the operation. The application only shows operations with the selected priority.

OP

This selection criterion refers to the operation number. The application only shows the operations with the specified operation number.

You can also use wildcards.

Tool

This selection criterion refers to the tool defined for the operation. The application shows all operations that are assigned to the specified tool.

You can also use wildcards.

Show split OPs

If you enable this checkbox, the application also shows split operations.

Customer name

This selection criterion refers to the customer name (designation) defined in the operation. The application shows all orders with the selected customer name (designation).

Cost center

This selection criterion refers to the cost center of the workplace defined for the operation. The application shows all operations that are assigned to workplaces of the selected cost center.

You can also use wildcards.

Planned start ... to ...

This selection criterion refers to the planned start date defined for the operation. The application only shows the operations whose planned start date is included in the selected period.

Scheduled start time ... to ...

This selection criterion refers to the scheduled start date of the operation. The application only shows the operations whose scheduled start date is included in the selected period.

Earliest start ... to ...

This selection criterion refers to the earliest start date of the operation. The application only shows the operations whose earliest start date is included in the selected period.

Latest end ... to ...

This selection criterion refers to the latest end date of the operation. The system only displays the operations whose latest end date is included in the selected period.

Actual start ... to ...

This selection criterion refers to the start date of the operation. The system only displays the operations whose start date is included in the selected period.

Actual end ... to ...

This selection criterion refers to the end date of the operation. The system only displays the operations whose end date is included in the selected period.



Use the MOC application "[Order overview](#)" to select operations using the actual dates of the order, i.e. the order start or order end.

Basic date start ... to ...

This selection criterion refers to the basic start date of the order. The application only displays the operations whose basic start date of the order is included in the selected period.

Basic date end ... to ...

This selection criterion refers to the basic end date of the order. The application only displays the operations whose basic end date of the order is included in the selected period.

Sales order

This selection criterion refers to the sales order defined in the order header. The application shows all operations that are assigned to the selected sales order.

Order index ... to ...

This selection criterion refers to the order index defined in the order header. The application displays all operations whose order index of the order header matches the selected order index.

Planned order

This selection criterion refers to the planned SAP order number that is defined in the order header.

Project number

This selection criterion refers to the project number of the operation. The application displays all orders that are assigned to the selected project number.

Order group

This selection criterion refers to the order group of the order header. The application displays all operations whose order group of the order header matches the selected order group.

You can also use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. The application shows all operations whose MRP controller of the order header matches the entered MRP controller.

You can also use wildcards.

MOP

Use this option to only show merged operations.

You can also use wildcards.

Show MOP

Use this option to specify the [merged operations](#) to be displayed.

By default, the application shows merged operations and individual operations. The application does not show the operations that are summarized in an MOP.

Show the operations logged on for each workplace

If you check this option, the application shows an operation multiple times if this OP is simultaneously logged on to multiple workplaces.

The category "logged on to workplace" (hidden by default) shows information about the workplace where the operation is logged on.

Check responsibility area

Using this option, the user can specify if the system checks the responsibility area of the workplace or the responsibility area of the object operation/order to display data. You need the following function authorizations to use this selection option: filterProdInd + chkresp

Detail application *Operations*

The table shows the below-mentioned fields, among other things. For further information refer to the following documents

- [Existing order data](#)
- [Current data and KPIs of the operation](#)

Status category

Status

The *status* column shows the bitmap ("LED") defined in the status configuration.

The totals line of this column displays the number of operations.

Status text

The current operation status specifies the status text.

Status since

Specifies the date and time since when the status has been set.

Predecessor status

Status of the preceding operation. This status specifies whether the preceding operation has already been started. This is important if you want to know if material needed for the current operation has already been processed or produced.

Secondary status

Displays the currently set secondary status.



Optionally, you can configure and use secondary statuses while customizing HYDRA.

Order category

Displays specific data for the [Operations](#) and [Orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol.

The glossary describes the standard order types. You can define additional order types as part of the system customization.

Category

The category summarizes similar order types. The following categories are available in the system:

FA = production order

PJ = project order

PM = maintenance order

KP = capacity order

GK = overhead order

Order

Shows the order number.

Sequence

Order sequence (only relevant if sequences are used).

OP

Shows the operation number.

Split

Shows the split number if the operation is a split operation.

Current number of splits

Shows the number of splits of an operation if the operation has been split.

Note

Shows the short text of the first note of an operation.

Printed

This column shows whether the time ticket has already been printed.

Specifications for production category

The relevant fields are:

Planned workplace

Workplace where the operation is planned.

Planned for

The "planned for" column shows whether the operation is planned

G for a group (pool of groups) or

M for a workplace (pool of workplaces).

Group

Group where the operation is planned.

OP dates category

Date specifications for the [Operation](#).

Remaining run time category**remaining run time**

Shows the remaining production time. The system uses a formula and different parameters to calculate this value. The formula is stored with the operation.

remaining run time

Shows the remaining production time. The system uses a formula and different parameters to calculate this value. The formula is stored with the operation.

Target times category

Target time specifications for the [Operation](#).

Additional, calculated fields:**Total setup time**

Sum total of setup time, additional setup time and retooling/teardown time.

Target execution time

Total of setup time plus processing time. **Primary quantity/secondary quantity/tertiary quantity category**

Target quantity

Quantity specifications for the [Operation](#).

Yield

Yield posted to the operation.

Scrap

Scrap posted to the operation.

Rework

Quantity to be reworked posted to the operation.

Problem quantity

Problem quantity posted to the operation.

Unit

Quantity unit of the values displayed.



The quantities listed here are displayed as base, primary, secondary and tertiary quantities. In general, you should only show one of these quantity types. The terminal collects quantities in the *primary quantity*.

Postings category**First logon**

Shows the date and time when the operation was logged on for the first time.

Last logoff

Shows the date and time when the operation was logged off the last time. This kind of information is useful if an operation was reactivated.

Date of last posting

Date of the last posting made for this operation.

Time of last posting

Time of the last posting made for this operation.

Logged on to workplace category

The below-mentioned fields pertaining to this category are only completed, if you set the option "Show the operations logged on for each workplace".

Workplace

Shows the workplace where the operation is currently logged on.

Short name

Short name of the workplace.

Name

Name of the workplace.

Group

Group of the workplace.

Cost center

Cost center of the workplace.

Toolbar

In general, the application takes over the parameters to call the function or target application from the table of the *Operations* detail application. For this reason, you should always select an operation before calling an application.

**Operation status (function authorization op.statchg)**

Function [Change operation status](#)

**Change secondary status (function authorization op.secstatchg)**

Function [Change the secondary status of an operation](#)

**Modify resource status (function authorization: op.resstatchg)**

Function [Change the resource status of an operation](#)

**Generate merged operation (function authorization: op.colopcreate)**

Function [Generate merged operation](#)

Use this function to combine the selected operations to form a merged operation. The merged operation is logged on to the terminal and the terminal shows this merged operation instead of the different single operations.

**Cancel merged operation (function authorization: op.coloprelease)**

Function [Cancel merged operation](#)

Use this function to cancel the merged operation. The application shows the different single operations.

**Order information (function authorization: orin)**

Use this button to call the application [Order information](#).

**Order overview (function authorization: orov)**

Use this button to call the application [Order overview](#).

**Schedule controlling: save baseline plan (function authorization: esvb)**

Function [Schedule controlling: save baseline plan](#)